

Using Q-SPECTOR

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Abstract: Q-SPECTOR is a Python program to analyse vibrational outputs from several mainstream computational chemistry software. Q-SPECTOR can build spectra based on the computed intensities (IR and VCD). Q-SPECTOR also generates PVDOS (Partial Vibrational Density of States) and *approximate* KED (Kinetic Energy Distribution) spectra based on cartesian displacements of the normal modes. The current version Q-SPECTOR v1.5 KED function detects bond stretching and valence angle bending motions, and torsional angle detection is in the development. The program has very flexible interface for PVDOS and KED atoms selection (based on atomic labels, weights, or elements). 'Stick' spectra are broadened using the pseudo-Voigt profile, with limiting cases of Gaussian or Lorentz broadening available as well. The current version has GAUSSIAN and AMBER vibrational output reading modules; support for ADF and JAGUAR is under the way.

I. Introduction ..no internal coordinates mess... in MM the force constants are defined, and in QC there are no bond stretch or angle bending force constants as such.

II. Partial Vibrational Density of States (PVDOS) Analysis.

PVDOS can be characterized via the corresponding mode composition factor $e_{j\alpha}^2$. By the definition, $e_{j\alpha}^2$ is equal to the $\langle T_{j\alpha} \rangle$ fraction of the ($\langle \dots \rangle$ averaged) kinetic energy stored in motion of atom j ($= 1, 2, \dots, N$) in the normal vibrational mode α ($= 1, 2, \dots, 3N - 6$):

$$e_{j\alpha}^2 = \frac{\langle T_{j\alpha} \rangle}{\sum \langle T_{i\alpha} \rangle} \quad (1)$$

The mode composition factor is related to the atomic masses m_j and normal mode displacement vectors $\mathbf{r}_{j\alpha}$ via the well-known formula:

$$e_{j\alpha}^2 = \frac{m_j \mathbf{r}_{j\alpha}^2}{\sum m_i \mathbf{r}_{i\alpha}^2} \quad (2)$$

The above equation 2 can be easily proven using the following three relations of the classic harmonic motion. First, the average kinetic energy $\langle T_{j\alpha} \rangle$ is equal to the average potential energy $\langle U_{j\alpha} \rangle$ according to the virial theorem, thus being a half of the full energy of the oscillator $E_{j\alpha}$ ($= \langle E_{j\alpha} \rangle$) :

$$\langle T_{j\alpha} \rangle = \langle U_{j\alpha} \rangle = \frac{1}{2} E_{j\alpha} \quad (3)$$

Second, the full energy of the oscillator $E_{j\alpha}$ is:

$$E_{j\alpha} = U_{j\alpha} \Big|_{T_{j\alpha}=0} = \frac{1}{2} k_{j\alpha} \mathbf{r}_{j\alpha}^2 \quad (4)$$

where $k_{j\alpha}$ is the force constant for the motion of atom j in mode α . Third, $k_{j\alpha}$ and m_j of atom j are connected via the frequency of the mode, f_α :

$$f_\alpha = \frac{1}{2\pi} \sqrt{\frac{k_{j\alpha}}{m_j}} \quad (5)$$

Collecting equations 1 and 3 through 5, we obtain the necessary relation 2 used to calculate PVDOS in Q-SPECTOR.

III. Kinetic Energy Distribution (KED) Analysis

Notably, atomic PVDOS considered above is essentially equivalent to the single atom KED. However, KED is normally considered as kinetic energy stored in a stretching, bending, or torsional vibration. Calculating kinetic energies associated with these types of oscillations would involve coupling displacements of at least two atoms. Consideration of a covalently bonded atom pair (either as a fragment of a large molecule or in a diatomic molecule) will be essential for the stretching and bending KED analysis following in subsections below.

III.a. Atom pairs. For the two atoms 1 and 2, the displacement vector $\mathbf{r}_{\text{CM}}$ of the Center of Mass of the 1-2 pair (CM) is given via formula (here, the normal mode index is omitted for clarity):

$$\mathbf{r}_{1-2} = \frac{m_1 \mathbf{r}_1 + m_2 \mathbf{r}_2}{m_1 + m_2} \quad (6)$$

Using 6, atom 1 and 2 displacement vectors $\mathbf{r}_{1-2}$ and $\mathbf{r}_{2-2}$ in the system moving with the center of mass of 1-2 pair are:

$$\begin{aligned} \mathbf{r}_{1-2} &= \mathbf{r}_1 - \mathbf{r}_{1-2} = \frac{m_2(\mathbf{r}_1 - \mathbf{r}_2)}{m_1 + m_2} \\ \mathbf{r}_{2-2} &= \mathbf{r}_2 - \mathbf{r}_{1-2} = \frac{m_1(\mathbf{r}_2 - \mathbf{r}_1)}{m_1 + m_2} \end{aligned} \quad (7)$$

Finally, according to equation 7 the collective KED factor associated with the two atoms oscillations relatively to the center of mass would be (cf equation 2):

$$m_1 \mathbf{r}_{1-2}^2 + m_2 \mathbf{r}_{2-2}^2 = \frac{m_1 m_2}{m_1 + m_2} \mathbf{r}_{12}^2 \quad (8)$$

where $\mathbf{r}_{12} = \mathbf{r}_1 - \mathbf{r}_2$

The importance of equation 8 will be obvious from the following subsections.

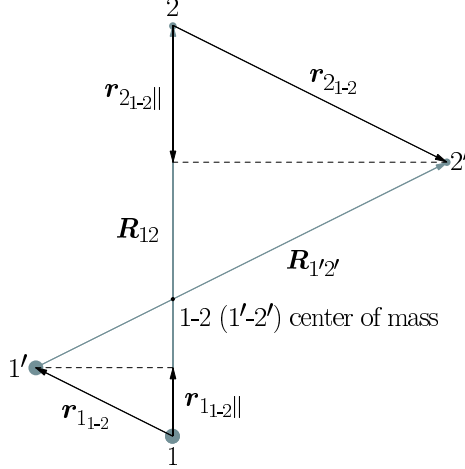


Figure 1: A schematic representation of $\mathbf{r}_{1-2\parallel}$ and $\mathbf{r}_{2-2\parallel}$ displacement vectors used to calculate a bond stretch KED in Q-SPECTOR. Vectors $\mathbf{R}_{12}$ and $\mathbf{R}_{1'2'}$ connect the two atoms in their equilibrium positions (points 1 and 2) and after the displacement is applied (points 1' and 2'). All the vectors are given relatively to the 1-2 (1'-2') center of mass reference system.

III.b. Valence Bond Stretching KED. Here we will do the following approximation for the bond stretching: the kinetic energy stored in 1-2 stretch vibration (assuming the two atoms 1 and 2 covalently bound) results from *projections* $\mathbf{r}_{1-2\parallel}$ and $\mathbf{r}_{2-2\parallel}$ of their displacement vectors $\mathbf{r}_{1-2}$ and $\mathbf{r}_{2-2}$ on the vector $\mathbf{R}_{12}$ connecting these two atoms equilibrium positions, see Figure 1. Notably, the displacements are considered relatively to the 1-2 pair center of mass. The bond stretch kinetic energy is then proportional to the sum of 1 and 2 kinetic energies (cf equation 2) resulting from the $\mathbf{r}_{1-2\parallel}$ and $\mathbf{r}_{2-2\parallel}$ displacements:

$$m_1 \mathbf{r}_{1-2\parallel}^2 + m_2 \mathbf{r}_{2-2\parallel}^2 = \frac{m_1 m_2}{m_1 + m_2} \mathbf{r}_{12\parallel}^2 \quad (9)$$

where $\mathbf{r}_{12\parallel}$ is the projection of $\mathbf{r}_1 - \mathbf{r}_2$ normal mode displacement vectors difference onto $\mathbf{R}_{12}$. The right side of equation 9 is derived using equations 7 and 8. Finally, using equations 2 and 9, 1-2 stretch KED composition factor $e_{12\alpha}^2$

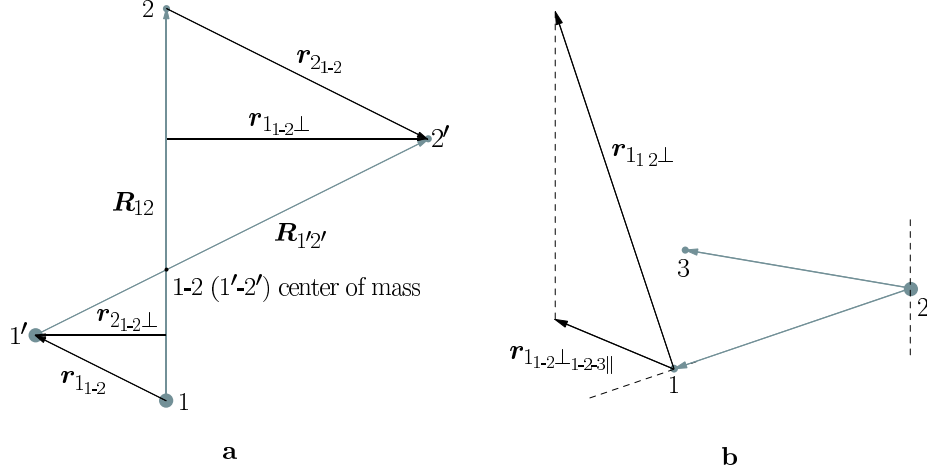


Figure 2: **a:** a schematic representation of $r_{1-2\parallel}$ and $r_{2-1\parallel}$ displacement vectors used to calculate valence and torsional angle bendings in Q-SPECTOR. **b:** a schematic representation of $r_{1-2\perp-1-2-3\parallel}$ displacement contribution to 1-2-3 angle bending.

for the mode index α is (the notation is analogous to $e_{j\alpha}^2$):

$$e_{12\alpha}^2 = \frac{\frac{m_1 m_2}{m_1 + m_2} r_{12\alpha}^2}{\sum m_i r_{i\alpha}^2} \quad (10)$$

Equation 10 is used to calculate 1-2 valence bond stretch KED in Q-SPECTOR. The list of $e_{12\alpha}^2$ values obtained for all α can be used to analyse where 1-2 stretch vibrations occur along the vibraional modes range, and what is their relative weight. To generate KED stretch spectra, Q-SPECTOR applies broadening to $e_{12\alpha}^2$ list. It is easy to show that equation 10 results in $e_{12}^2 = 1$ for a diatomic molecule vibrational mode, using $r_{12\parallel} = r_{12}$ and $r_2 = -\frac{m_1}{m_2} r_1$. This reflects proper behavior of formula 10 in case when 100% of the mode vibrational energy is stored in 1-2 stretch.

III.b. Valence Angle Bending KED. Describing 1-2-3 valence angle bending, where 1-2 and 2-3 atom pairs are assumed to be bound, we first naturally suppose that bending (in contrast to stretching, see above) comes from the displacements, perpendicular to the 1-2 and 2-3 bonding vectors R_{21} and R_{23} .

These normal displacements will be denoted $\mathbf{r}_{1-2\perp}$ and $\mathbf{r}_{2-3\perp}$ for 1-2 pair, see Figure 2, and $\mathbf{r}_{2-3\perp}$ and $\mathbf{r}_{3-2\perp}$ for 2-3 pair. Notably,

$$\mathbf{r}_{i-i-j} = \mathbf{r}_{i-i-j\parallel} + \mathbf{r}_{i-i-j\perp} \quad (11)$$

1-2-3 valence angle bending is also limited to the displacements which occur within the plane formed by the three atoms 1-2-3. One more The second step of projection should be done, removing those components of displacements which are normal to the 1-2-3 plane (the normal components can be used for the description of torsional contributions to KED, see below). Following our notation scheme, $\mathbf{r}_{1-2\perp}$ when projected onto 1-2-3 plane would be called $\mathbf{r}_{1-2\perp\perp 1-2-3\parallel}$, see Figure 2. Then the amount of kinetic energy stored in 1-2-3 angle bending motion can be approximated as sum of 1-2 and 2-3 'bending' components (cf. equation 9), given below:

$$\begin{aligned} m_1 \mathbf{r}_{1-2\perp\perp 1-2-3\parallel}^2 + m_2 \mathbf{r}_{2-3\perp\perp 1-2-3\parallel}^2 &= \frac{m_1 m_2}{m_1 + m_2} \mathbf{r}_{12\perp\perp 1-2-3\parallel}^2 \\ m_2 \mathbf{r}_{2-3\perp\perp 1-2-3\parallel}^2 + m_3 \mathbf{r}_{3-2\perp\perp 1-2-3\parallel}^2 &= \frac{m_2 m_3}{m_2 + m_3} \mathbf{r}_{23\perp\perp 1-2-3\parallel}^2 \end{aligned} \quad (12)$$

where $\mathbf{r}_{12\perp\perp 1-2-3\parallel}$ is the component of $\mathbf{r}_1 - \mathbf{r}_2$ normal mode displacement vectors difference normal to $\mathbf{R}_{12}$, further projected onto the 1-2-3 plane. For mode index α , the fraction of kinetic energy stored in 1-2-3 angle bending then can be written as follows :

$$e_{123\alpha}^2 = \frac{\frac{m_1 m_2}{m_1 + m_2} \mathbf{r}_{12\alpha\perp\perp 1-2-3\parallel}^2 + \frac{m_2 m_3}{m_2 + m_3} \mathbf{r}_{23\alpha\perp\perp 1-2-3\parallel}^2}{\sum m_i \mathbf{r}_{i\alpha}^2} \quad (13)$$

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IV. Q-SPECTOR Input File and Usage.

The input (instructions) file format is as follows:

```
&output
spectra [IR,VCD,PVDOS,KEDSTRETCH,KEDBEND]
shape [SIGMA GAMMA]
range START STOP
resolution RES
&
&selection
element1 [E1_1 E1_2 ...]
element2 [E2_1 E2_2 ...]
element3 [E3_1 E3_2 ...]
weight1 [W1_1 W1_2 ...]
weight2 [W2_1 W2_2 ...]
weight3 [W3_1 W3_2 ...]
label1 [L1_1 L1_2 ...]
label2 [L2_1 L2_2 ...]
label3 [L3_1 L3_2 ...]
cutoff CUTOFF
&
```

...the above format is to be explained...

Using Q-SPECTOR on input.qsp with GAUSSIAN outputfile :

```
>Q-Spector-1.5.4.py GAUSSIAN input.qsp output.g03out output.dat
```

which results in output.dat spectrum data file. The output data has two column format, where the first column is the wavenumber (cm^{-1}) and the second is intensity.

Appendix

Exact formulae for projected displacement differences $\mathbf{r}_{12\parallel}$ and $\mathbf{r}_{12\perp_{1-2-3\parallel}}$ used to calculate $e_{12\alpha}^2$ (equation 10) and $e_{123\alpha}^2$ (equation 13), correspondingly.

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